PERFORMANCE ENHANCEMENT OF SOLAR WATER PUMP FOR IRRIGATION APPLICATION

A PROJECT REPORT PHASE II

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

A solar photovoltaic water pumping system of 250 W power capacity was developed and investigated its performance under the influence of panel cooling using active heat sinks. The experimental observations have been made under the climatic conditions of Coimbatore city in India during the year 2015. The performance was evaluated under five different modes such as: (a) photovoltaic panel without cooling; (b) photovoltaic panel with air cooling over the panel surface; (c) photovoltaic panel with air cooling at the beneath of the panel; (d) photovoltaic panel with water cooling over the panel and (e) photovoltaic panel cooling with water at the beneath of the panel. The performance characteristics of solar photovoltaic water pumping system such as, photovoltaic cell temperature; photovoltaic power output; photovoltaic electrical efficiency, motor characteristics and pump discharge with reference to different ambient conditions are evaluated and compared.

The panel cooling using air and water has significantly reduced the cell temperature and maintain in the range between 25 degree C and 40 degree C. The experimental results confirmed that water cooling is more effective when compared to air cooling due to its higher specific heat. The cooling of photovoltaic panel over the beneath of the panel has significant influence on reducing the cell temperature and enhance the panel output. The cooling of photovoltaic panel using air has been improved the photovoltaic efficiency, subsystem efficiency (pump discharge efficiency) and total efficiency by about 3.5%, 2.5% and 1.8%, respectively. The cooling of panel using water has improved the photovoltaic efficiency, subsystem efficiency (pump discharge efficiency) and total efficiency by about 4.2%, 3.2% and 2.1 %, respectively when compared to the system without cooling. The experimental results confirmed that, the performance enhancement of photovoltaic water pumping system under the influence of panel cooling is technically feasible.

The economic analysis of the systems confirmed that solar photovoltaic water pumping system has 6.4 years payback period for the considered operating conditions. After the payback period, the pump can be operated without running cost. The return on

investment and net present value assessments confirmed that, investment in solar photovoltaic water pumping system is an economically feasible option. Moreover, the environment impact assessment results for the period of ten years confirmed that, the solar photovoltaic water pumping system has significant reduction in carbon-dioxide emissions due to the direct conversion of solar energy. The environmental impact assessments confirmed that, solar photovoltaic water pumping system is an environment friendly option. Moreover, the solar photovoltaic water pumping system is identified as an energy efficient option due to its improved performance of photovoltaic panel. Hence, it is recommended to use the solar photovoltaic water pumping system in domestic, industrial, commercial and agricultural sectors.

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CHAPTER 1

INTRODUCTION

The fast depletion of conventional fossil energy sources and its unfavourable environmental impacts have created research interest in choosing a renewable energy system to power the water pumping system. The optimum utilization of water and energy resources has more essential in the future. In India, the potential availability of renewable energy sources such as, solar photovoltaic, solar thermal, wind, biomass and its various hybrid forms provides good solutions to the energy related issues. Solar energy is used in water pumping systems either in the form of photovoltaic electric power out to power the electric motors or in the form of solar thermal energy harvesting through solar thermal collectors to power either Rankine cycle (or) Brayton cycle (or) Stirling cycle. Out of these two options, solar photovoltaic is the viable option for water pumping applications due to its simplicity, economical and easy to install with less maintenance.

In India, about 58% of the land area are potentially be solar hot spots with 5 kWh/m2/day of annual average global isolation. The solar energy availability could fulfil the increasing power requirements specifically for agriculture water pumping applications in a decentralized, efficient and sustainable manner in the form of solar photovoltaic energy conversion systems. The performance of photovoltaic water pumping systems depends on the ambient parameters such as,solar intensity, ambient temperature, ambient relative humidity and ambient wind velocity. The renewable energy water pumping technologies in India are influenced by economic factors such as, financial incentive schemes provided by the central and state governments.

To Enhance solar water pump system is essential for an electrical pump system in which the electricity is provided by one or several Photovoltaic (PV) panels.

1.1 RENEWABLE ENERGY WATER PUMPING SYSTEM

In India, electrical and diesel-powered water pumping systems are widely used in agricultural applications. However, connecting the grid electricity with electrical water pumping systems in remote locations is more expensive. The use of diesel power plants for water pumping applications are having negative impacts on environment and depends on fossil fuel, which are getting depleted drastically. To meet the energy demand and to reduce the environmental impacts for water pumping applications, especially in agricultural sector, the idea of integrating renewable energy technologies such as solar photovoltaic (Pytlinski, 1978; Sontake et al. 2016), with water pumps has been proposed by many researchers around the world. The photovoltaic water pumping systems are having less maintenance with minimum losses and negligible carbon-dioxide emissions when compared to other conventional technologies. Earlier reported review studies confirmed that solar photovoltaic water pumping systems are having highest utilization in agriculture sector (Chandel et al. 2015; Rawat et al. 2016).

1.2 BACKGROUND OF THE RESEARCH WORK

In India has approximately 58% of the geographical area can potentially be solar hot spots in the country with more than 5 kWh/m^2/day of annual average global isolation (Ramachandra et al. 2011). The solar energy availability in India could fulfil the increasing power requirements in a decentralized, efficient and sustainable manner in the form of solar photovoltaic energy conversion systems. The performance of photovoltaic water pumping systems depends on the ambient parameters such as, solar intensity, ambient temperature, ambient relative humidity and ambient wind velocity. The renewable energy water pumping technologies in India are influenced by many techno-economic factors including the financial incentive schemes providedby the central and state governments.

1.3 SCOPE OF THE RESEARCH

Photovoltaic energy conversion is one of the best ways to harvest the solar energy. Many researchers around the world have investigated the performance of solar photovoltaic water pumping systems. However, the instantaneous performance of the photovoltaic water pumping system was highly influenced by ambient parameters such as solar intensity, wind velocity and ambient temperature. In order to enhance the performance of the system, a simple active photovoltaic panel cooling system is proposed in this research work to enhance the performance of the pumping system. Four cooling modes are proposed in this research work namely: (a) cooling of photovoltaic panels using water sprayed above the panel; (b) cooling of photovoltaic panel using water sprayed below the panel; (c) cooling of photovoltaic panel using air passed over the photovoltaic panel and (d) cooling of photovoltaic panels using air passed below the photovoltaic panel. The performance of photovoltaic water pumping systems working with four modes is compared against the photovoltaic pumping system without cooling.

1.4 OBJECTIVES AND METHODOLOGY

The following objectives are formulated in this research work:

- To develop a solar photovoltaic water pumping test facility with necessary instrumentations.
- To investigate its energy performance at Coimbatore climatic conditions (latitude10.98^{o}N and longitude 76.96^{o}E).
- To evaluate the technical, economical and environmental benefits for water pumping applications.

- To develop a mathematical model for predicting the energy performance under different climatic conditions.
- To evaluate its feasibility for irrigation applications.

The research investigation has been carried out in the following approach:

- Review of studies related to the photovoltaic water pumping systems.
- Problem was identified based on the literature review and references.
- Experiment are carried out under various kind of climatic conditions.

1.5 STRUCTURE OF PROJECT

This project report includes nine chapters. The **Chapter 1- Introduction** presents an overview of renewable energy water pumping system, background of solar photovoltaic water pumping systems, scope of the research, objectives and methodology adopted in the present research work.Finally, an overview of thesis work is descried.

In **Chapter 2-Literature Survey,** a comprehensive review on studies reported with solar photovoltaic water pumping system have been presented. Moreover, the studies reported with cooling of photovoltaic panels are presented. Based on the literature review, the problem was identified and possible solutions are described

In **Chapter 3 SOLAR WATER PUMP,** the equations used for predicting the performance of a solar photovoltaic water pumping system are presented. The assumptions made in the analysis arealso described.

In **Chapter 4 PV PANEL**, the detail description on experimental setup, instrumentation and experimental procedure are described. Additionally, the uncertainties in the experiments are presented.

In **Chapter 5 ECONOMICAL ASSESSMENT**, the results obtained in experiments are presented. The results predicted using mathematical models are compared with experimental data. Furthermore, the performance of the solar photovoltaic water pumping system under five different modes is compared.

In **Chapter 6 ASSEMBLING OF COMPONENTS**, the results obtained in experiments are presented. The results predicted using mathematical models are compared with experimental data. Furthermore, the performance of the solar photovoltaic water pumping system under five different modes is compared.

In **Chapter 7 ADVANTAGES AND DISADVANTAGES OF SOLAR WATER PUMP**, the results obtained in experiments are presented. The results predicted using mathematical models are compared with experimental data. Furthermore, the performance of the solar photovoltaic water pumping system under five different modes is compared.

In **Chapter 8- SCOPE OF FURTHER RESEARCH**, After the reference test, the performance was carried out using air as cooling medium on the above and the beneath surface of the panel.

In **Chapter 9- Conclusion** Finally, the major conclusions arrived in this research are consolidated .The scope for further research is also presented.

1.6 SUMMARY

An overview of solar photovoltaic water pumping system was described in this chapter. In addition, background of the solar photovoltaic water pumping system, the scope of the research, objectives and methodology were presented. The content of the thesis were described at the end of the thesis. The brief summary of reported research and developments on the topic of investigation are reviewed in Chapter-2 as Literature Review.

CHAPTER 2

LITERATURE REVIEW

This chapter deals with the literature survey of various patents available in the market is as follows.

TITLE 1: Performance evaluation of water pumping system powered by solar energy

Author Name:

M. Mohanraj

P. Chandramohan M.Sakthivel

Dated: 23 Nov 2019

In this paper, the performance of a photovoltaic water pumping system was experimentally investigated under the influence of panel cooling using air and water as working fluids. The experimental observations have been made under the climatic conditions of Coimbatore city in India during the year 2017. The system performance was evaluated in four different cooling modes such as: (a) with air cooling over the top surface of the panel; (b) with air cooling over the bottom surface of the panel; (c) with water cooling over the top surface of the panel and (d) with water cooling over the bottom surface of the panel. The performances of a photovoltaic water pumping system working with four different panel cooling modes are compared with the system without panel cooling. The economical and environmental impacts have been assessed for a period of twenty years. The results showed that, the water cooling provided over the bottom surface of the panel has significant performance enhancement when compared to other cooling modes. The economic assessment results showed that, the photovoltaic water pumping systems

using water cooling over the bottom surface of the panel has payback period of 7.3 years with 13.7% return on investment and 10.9% internal rate of return. The trial experiments confirmed that, it is possible to reduce the carbon-dioxide emissions by about 20.4 tons during the period of twenty years life when compared to the grid connected water pumping systems powered by coal based power plants.

TITLE 2: Performance of photovoltaic water pumping systems under the influence of panel cooling.

Author Name:

Kamaruzzaman

Sopian

Lower operating temperatures of the photovoltaic (PV) cells increase the performance and efficiency of any PV installation. The efficiency of solar photovoltaic water pumping system (SPVWPS) decreases considerably with the increase in the PV cell temperature. In this paper, the performance of a 2 hp SPVWPS has been investigated experimentally, for the influence of panel cooling, using water. The experimental observations have been made under climatic conditions of Visvesvaraya National Institute of Technology, Nagpur campus, India, during the year 2018. The performance was evaluated under four different cases: (a) without panel cooling, (b) with water cooling on the top of the panel surface, (c) with water cooling on beneath the surface of the panel, and (d) with water cooling beneath the surface of the panel using jute. The effect of different cooling cases on the various performance parameters such as discharge, power output, pump efficiency, and system efficiency has been analyzed and discussed. The results showed that the water cooling on the top of the panel and beneath the surface of the panel with jute has considerable influence on performance enhancement when compared with other cases.

TITLE 3: Solar powered water pumping system.

Author Name:

Tamer Khatib

Pradeep Arya

Water pumping is one of the most widespread technological processes, and they are mostly used for supplying water to houses, private residents, farms, agricultural irrigation, etc. In modern conditions characterized by a lack of fossil fuels and increased attention to environmental protection, solar water pumping systems (SWPS) have become one of the most common applications of solar photovoltaic, especially in developing countries, for domestic applications in remote locations [1,2]. This is due to its ability to fulfil a wide range of technological needs, the simplicity of its technical implementation, its high reliability and its cost effectiveness.

TITLE 4: Development of solar powered irrigation system development of solar powered irrigation system

Author Name:

Sami Eusuf

A solar powered pumping system methods needs to take account of the fact that demand for irrigation system water will vary throughout the year. Peak demand during the irrigation system seasons is often more than twice the average demand. This means that solar pumps for irrigation are under-utilized for most of the year. Attention should be paid to the system of irrigation water distribution and application to the crops. The irrigation pump system should minimize water losses, without imposing significant additional head on the irrigation pumping system.

There are several technology alternatives for supplying power or lift to groundwater systems including wind turbines, windmills, generators, solar arrays, and hand powered pumps. The main driving factors for selecting the appropriate technology are regional feasibility, water demand, system efficiencies, and initial and long-term costs. Other factors often include the need for power and water reserves in the form of batteries and storage tanks. Solar-powered systems are often considered for use in developing countries instead of other forms of alternative energy because they are durable and exhibit long-term economic benefits. Solar powered water pumping has been recognized as suitable solution for grid- isolated rural locations in poor countries where there are high levels of solar radiation. Solar powered water pumping systems can provide drinking water without the need for any kind of fuel or the extensive maintenance required by diesel pumps. SPPS is also not adequate , but can work for small-scale drip irrigation systems. A large-scale SPPS can be considered one that serves over 240 peoples. Photovoltaic (PV) solar panels are often used for agricultural operations, especially in remote areas or where the use of an alternative energy source is desired. In particular, they have been demonstrated time and time again to reliably produce sufficient electricity directly from solar radiation to power livestock and irrigation watering systems. Solar water pumps may be especially useful in small scale or community basedirrigation, as large scale irrigation requires large volumes of water that in turn require a large solar PV array. As the water may only be required during some parts of the year, a large PV array would provide excess energy that is not necessarily required, thus making the system inefficient.Solar PV water pumping systems are used for irrigation and drinking water India. The majority of the pumps are fitted with a 200 watt - 3,000 watt motor that receives energy from a 1,800 Wp PV array. The larger systems can deliver about 140,000 liters of water/day from a total head of 10 meters. The photovoltaic panels are to be cooled uniformly to improve its overall performance. The series of losses occurred in the photovoltaic panel are due to geometry of the concentrator, losses due to optics, reflection and tracking and non-uniform illumination. These losses will influence cell temperature and increases the series resistance, which lowers the overall efficiency. Non uniform cooling of photovoltaic panels will results in decrease in cell temperature, increase in resistance, decrease in fill factor and decrease in conversion efficiency. The concept of heat pipe was proposed by many researchers to cool the photovoltaic panels.

Akbarzadeh and Wadowski (1996) investigated the effectiveness of the heat pipe assisted cooling system for photovoltaic systems. It was reported that the temperature of the panel was reduced to 46 °C from 84 °C with a heat pipe efficiency of 94%. Moreover, the heat pipe assisted photovoltaic panel cooling is getting more attention. Photovoltaic power output of the panel was significantly improved by integrating a heat pipe with photovoltaic panels to enhance its performance (Siecker et al. 2017).

In a similar approach, Benlarbi et al. (2004) presented an on-line fuzzy optimization of a solar photovoltaic water pumping system driven by three different motors such as an excited DC motor, a permanent magnet synchronous motor, or an induction motor coupled to a centrifugal pump. The fuzzy optimization procedure aims to maximize the global efficiency by increasing the drive speed and the water discharge rate of the coupled centrifugal pump. They confirmed the suitability of using the fuzzy logic procedure as a standard optimization algorithm for photovoltaic water pumping drives. They also reported that a permanent magnet synchronous motor is a good choice for the solar photovoltaic water pumping system drive because DC motors require periodical maintenance. However, an induction motor is technically a competitive choice. Similarly, Fernandez-Ramoset al. (2010) improved the performance of solar photovoltaic water pumping systems by using a standard frequency converter and PLCs. It has been proved that the addition of a standard PLC to a solar photovoltaic water pumping system based on standard frequency converters will avoid the stopping of the system during sudden decreases of the solar radiation. In another work, Benghanem and Arab (2007) reported the performance of a domestic solar photovoltaic water pumping system installed in remote areas of Algeria. Due to high cost of the setup, the maintenance and the performance monitoring of a large number of solar photovoltaic water pumping system, they developed a universal data acquisition system for monitoring the performance and optimizing the size of solar photovoltaic water pumping systems located in different remote regions of Algeria.

The detailed open literatures presented on solar photovoltaic water pumping systems confirmed that many research and development investigations have been made on solar photovoltaic water pumping systems. However, the performance of solar photovoltaic water

pumping systems under the influence of air and water cooling is not reported in open literature. Cooling of photovoltaic panels is one of the possible solutions to improve the efficiency of photovoltaic water pumping systems. Hence, an attempt has been made in this research work to explore the possibility of improving overall performance of photovoltaic water pumping using air and water as cooling medium. A mathematical model has been developed for predicting the performance of solar photovoltaic water pumping systems and validated with experimental results. The detailed research background on solar photovoltaic water pumping system was reviewed in this chapter. Based on literature review, the research problem and the possible solutions are identified and described. The thermodynamic modelling of solar photovoltaic water pumping is discussed in the following chapter titled “Modelling of solar photovoltaic water pumping systems”.

CHAPTER 3

MODELING OF SOLAR PHOTOVOLTAIC WATER PUMPING SYSTEMS

The solar photovoltaic water pumping system consists of solar photovoltaic panels, inverter, variable frequency drive, AC induction motor and a centrifugal pump. The mathematical equations used for predicting the performance of each component are described in this chapter. This chapter is subdivided into following major subsections:

(i). Modeling of an inverter
(ii). Modeling of an electrical motor
(iii). Modeling of a centrifugal pump
(iv). Modeling of an overall system
(v). Analysis of variance
(vi). Economical analysis
(vii). Environmental impact assessments

Finally the summary of this chapter is presented. The solar photovoltaic water pumping systems consists of DC-DC converter in power conditioning unit with maximum power point tracking facility, DC-AC inverter and centrifugal pump assisted by an AC induction motor. The performance of the solar photovoltaic water pumping systems depends on ambient parameters such as, solar intensity, ambient temperature, ambient wind velocity and pump operating characteristics such as head and speed of the pump. The photovoltaic laminate consists of tempered glass cover with maximum light transmission and minimum heat absorption. The energy flow diagram of a solar photovoltaic water pumping system is illustrated in Figure 3.1.

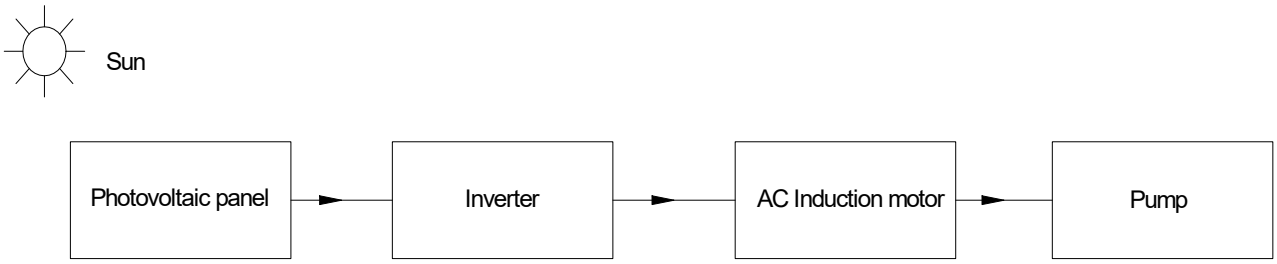


FIG 3.1 modelling of solar photovoltaic water pumping systems

PRINCIPLES

For any solar pumping system, the capacity to pump water is a function of three main variables pressure, flow, and power to the pump.

3.1 PRESSURE

For purposes of designing a solar pumping system, pressure can be thought of as the work that thepump must overcome to move a certain amount of water. This is most often expressed in either feet of head or psi (pounds per square inch). This is also referred to as pressure loss.

3.2 FLOW

The amount of water that a system can move during a given time period. It is usually talked aboutin terms of gallons per minute (GPM) or gallons per hour. If everything else remains the same, flow will decrease when pressure increases and vice versa.

3.3 POWER TO THE PUMP

Every solar water pump can produce a range of flows and pressures. Solar pumps draw a certain amount of power according to the amount of pressure that needs to be produced to deliver the water. Power is expressed in Watts, and PV panels are rated in Watts. When sizing a

photovoltaic panel array, it makes sense to supply the amount of power that is needed. Adding more PV than is needed might enable the pump to turn on earlier and later in the day or under low-light conditions, but having extra PV power might not increase the flow rate when the sun is shining in full.

MAIN SOLAR WATER SYSTEM COMPONENTS

THE SOURCE

Water sources can be deep drilled wells, streams, springs, ponds, rivers, etc.The main variables, in addition to initial cost, that factor into system design are the recharge rate of the source and the volume of the source reservoir. Ideally, the source should recharge faster than pump can take water out of it. For example, if the pump will produce three gallons per minute, the source should be able to recharge at a rate that is equal to or greater than three gallons per minute. If the pump takes water away faster than the source can recharge, the reservoir can run dry, the pump will then be run dry, and that should be avoided to prevent damage to the pump.

THE PUMP

This is the heart and soul of the solar water pumping system. Pumps fall along a spectrum of high- flow/low-head to low- flow/high head. Solar pumps are rated according to the voltage of electricity that should be supplied. A 12 volt pump is a small one, 24 volt is more the norm, while 48 volts and upwards will require more power and might pump more water.

Smaller wire sizes can be used in higher voltage systems without sacrificing power output from panels to pump. Some pumps require certain accessories to function optimally. These include filters, float valves, switches, etc.

1. PHOTOVOLTAIC (PV) PANELS

The photovoltaic panels make up most (up to 80%) of the systems cost. The size of the PV- systemis directly dependent on the size of the pump, the amount of water that is required (m^3/d) and the solar irradiance available. A panel is rated in watts of power it can produce. The SPV water pumping system should be operated with a PV array capacity in the range of 200 Watts peak to 5000 Watts peak, measured under Standard Test Conditions (STC). Sufficient number of modules in series and parallel could be used to obtain the required PV array power output. The power output of individual PV modules used in the PV array, under STC, should be a minimum of 74 Watts peak, with adequate provision for measurement tolerances. Use of PV modules with higher power output is preferred. Indigenously produced PV module(s) containing mono/ multi crystalline silicon solar cells4.

2. MOTOR PUMP-SET

Following types of motor pump sets could be used in the SPV water pumping systems

- Surface mounted DC motor pump-set
- Submersible DC motor pump set
- Submersible AC motor pump set
- Floating DC motor pump set5.

3. MOUNTING STRUCTURES AND TRACKING SYSTEM

To enhance the performance of SPV water pumping systems, it is desirable to use a tracking system. Manual, passive and auto tracking are permitted. The PV modules will be

mounted on metallic structures of adequate strength and appropriate design, which can withstand load of modules and high wind velocities up to 150 km per hour. The support structure used in the pumping system will be hot dip galvanized iron (G.I). The G.I. structures for mounting the Solar panels could be so designed, that these can be manually/ auto adjusted for optimal tilt throughout the year.

4. CONTROLLER

The purpose of the controller is twofold. Firstly, it matches the output power that the pump receives with the input power available from the solar panels. Secondly, a controller usually provides a low voltage protection, whereby the system is switched off, if the voltage is too low or too high for the operating voltage range of the pump. This increases the lifetime of the pump thus reducing the need for maintenance.

5. INVERTER

It changes the direct current from the solar panels into alternating current for the pump. The supported power range of inverters extends from 0.15 to 55 kW and can be used for larger irrigation systems. However, the panel and inverters must be sized accordingly to accommodate the inrush characteristic of an AC motor.

6. THE WATER DISTRIBUTION SYSTEM

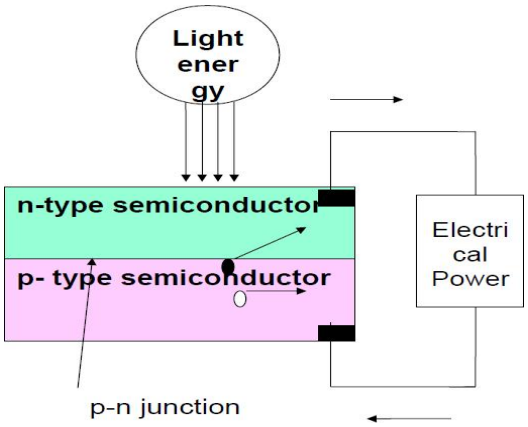
In many countries of the world, trenches are dug to gravity feed water through an intricate network of irrigated plots or holding tanks. Here farmers tend to use black polyethylene piping. Whateverthe system is to get water from A to B, there important thing to consider is the smaller the diameter piping and the longer the piping run, the harder a pump has to work and the more pressure the pump must create.

CHAPTER 4

PV PANEL

4.1 PHOTOVOLTAIC EFFECT

Photovoltaic energy conversion is one of the best ways to convert the solar energy to electricity. Solar cell is also called as photo galvanic cell. Photo galvanic cell converts the solar energy into electrical energy. Solar cells working on the principles of photovoltaic effect. The devices based on this effect is called photovoltaic device. Solar cell consists of a n-type semiconductor (emitter) and p-type semiconductor (base). These two layers are sandwiched and form p -n junction. The surface is coated with anti-reflection coating to avoid the loss of incident light due to reflection.



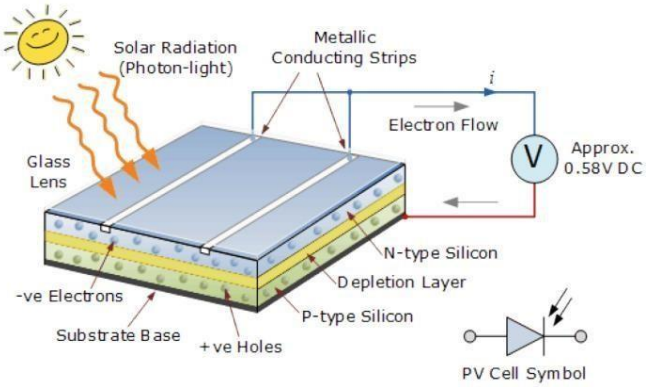


FIG 4.0 & FIG 4.1 photovoltaic effect

4.2 PHOTOVOLTAIC CELLS

Photovoltaic panels are made of several layers of silicones. Silicon is a semi-conducting material occurs in following forms: Amorphous Silicon. Polycrystalline (or Multi-crystalline) Silicon Mono-crystalline Silicon. These materials are used for different photovoltaic panels with different characteristics and outputs. First generation solar cells are mainly made out of polycrystalline or mono-crystalline silicon. Second generation silicones consist of amorphous material such as copper indium gallium selenide (CIGS), cadmium telluride (CdTe), gallium arsenide(Ga As) and amorphous silicon (a -Si:H). Third-generation cells are not commercially available. It includes organic photovoltaics, copper zinc tin sulphide, perovskite solar cells, dye- sensitized solar cells and quantum dot solar cells. Third-generation technologies aim to enhance the electrical performance of second-generation thin-film technologies, with low production costs. Current research is targeting conversion efficiencies of 30–60%, while retaining low-cost materials and manufacturing techniques.

4.3 WORKING OF PHOTOVOLTAIC PANEL

When a solar panel exposed to sunlight, the light energies are absorbed by semi- conductor materials. Due to the absorbed energy, the electors are liberated and produce the external direct(DC) current. Generated DC current is converted into 240 volt AC current using inverter. When p and n layers are connected to an external circuit, the electors flow from n layer to p layer, which results in current generation. Electors leave the solar cell as current give up their energy to what evet is connected to the solar cell.

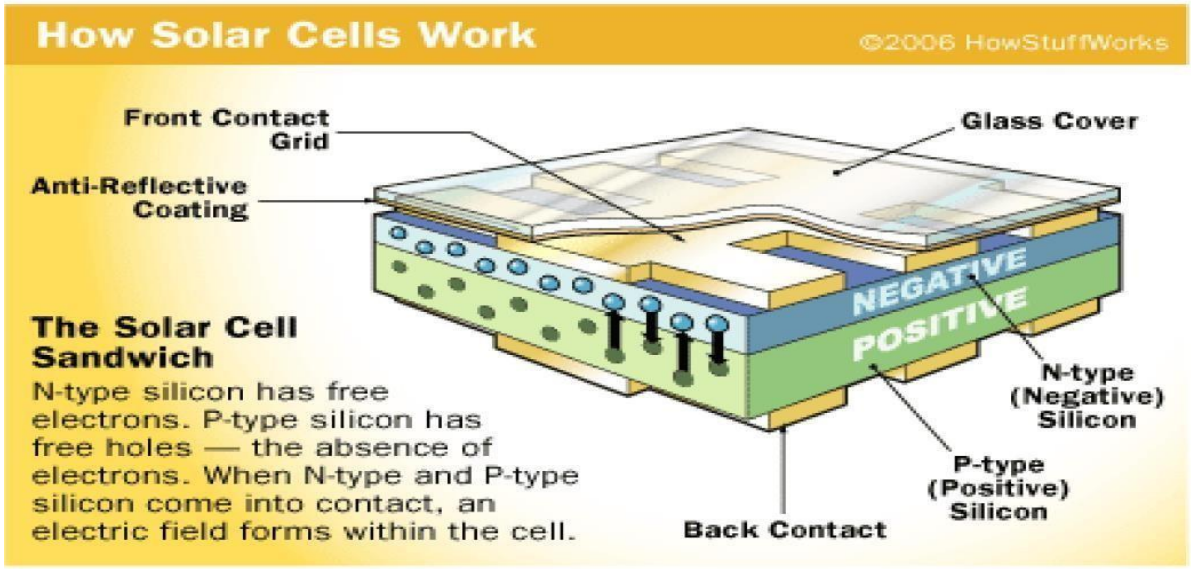


FIG 4.3 working of photovoltaic panel

4.4 PHOTOVOLTAIC WATER PUMPING SYSTEMS

SPWPSs consist of solar photovoltaic panels, a motor and a pump, which is depicted Figures. Depending on the system design, it require storage batteries and a charge regulator. The motor is chosen according to the power requirement and the type of current output of the system. If the motor uses alternative current(AC), it is necessary to install a direct current (DC) to AC converter. Battery-less SPWPSs are low cost, which requires less maintenance compared to battery powered systems. However, the storage batteries have the advantage of providing consistent performance during lean and off sun shine hours.

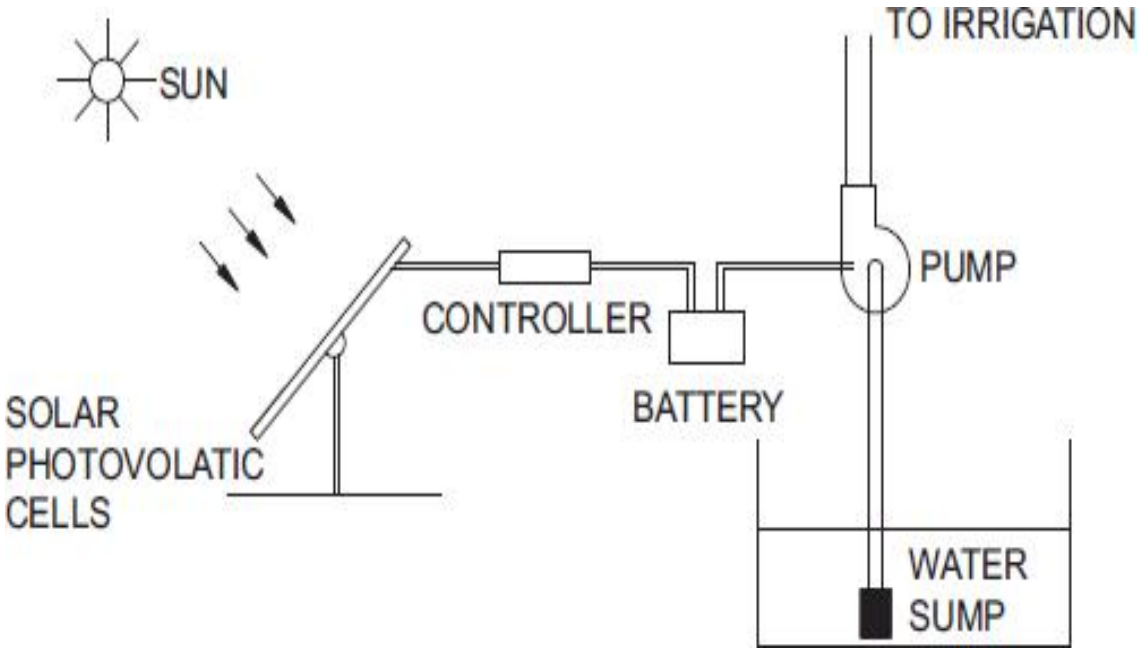


FIG 4.4 photovoltaic water pumping systems

FIG 4.5 image of solar panel and pump

PHOTOVOLTAIC SPECIFICATIONS

Current at P_{max}	7.85 Amps
Open circuit voltage V_{oc}	43.99V
Short circuit current I_{sc}	8.39 Amps
Temperature co efficient of P_{max}	-0.41%˚C
Temperature co efficient of I_{sc}	0.04
Temperature co efficient of V_{oc}	-32% ˚C
Cell type	Multi crystalline
Cell size	156mm x 156mm
Module dimension	1975 mm (L) x 988 mm (W) 50mm (t)
Solar irradiance (G)	1000 W/m^2
Area of the Panel	6 m^2

TABLE 1.1

WATERPUMP SPECIFICATION

Voltage	230 Volts
Frequency	50 Hz
Current	2.5 Amps
Speed	2800 rpm
Power	0.5 HP and 0.375 kw
Pump	Surface Pump (Self priming

TABLE 1.2

Type of motor	Single phase A.C motor
Voltage, Current	230 Volts, 2.5 Amps
Frequency / Speed	50 Hz / 2800 rpm
Power	0.5 HP and 0.375 kW
Pump	Centrifugal pump
Suction pipe size	25.4 mm
Delivery pipe size	25.4 mm

TABLE 1.3

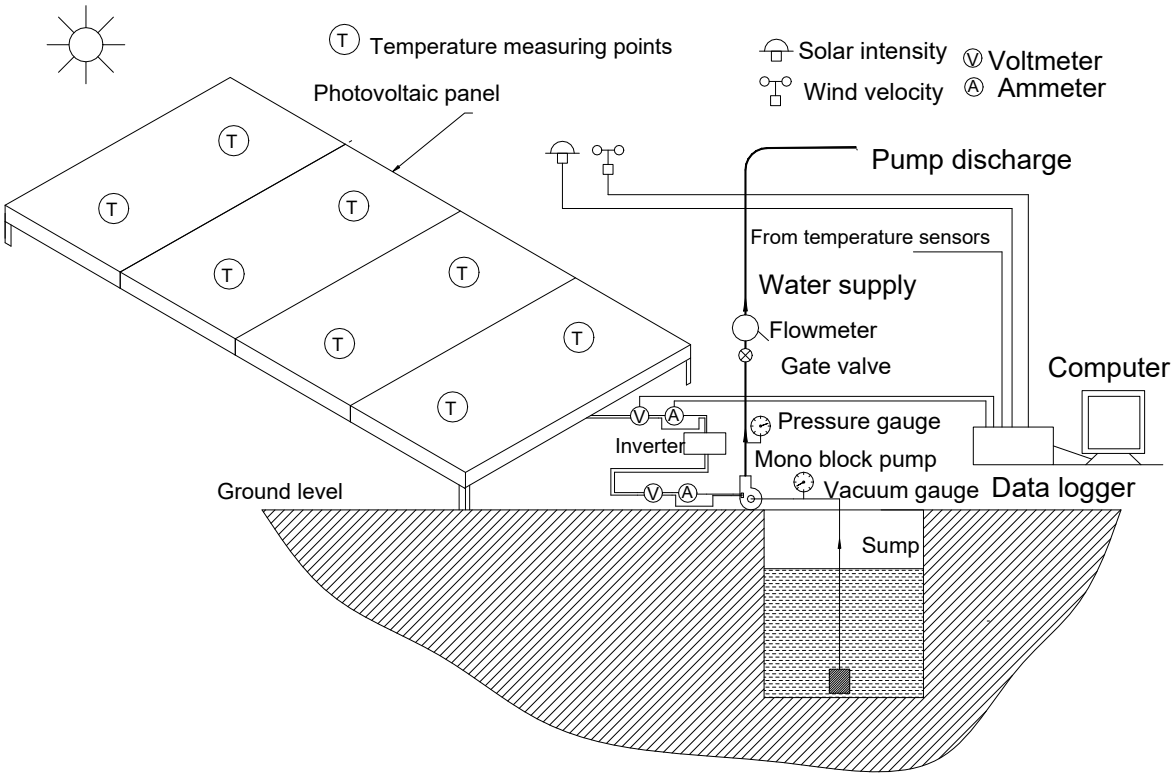


FIG 4.6 Schematic diagram of experimental setup with instrumentation

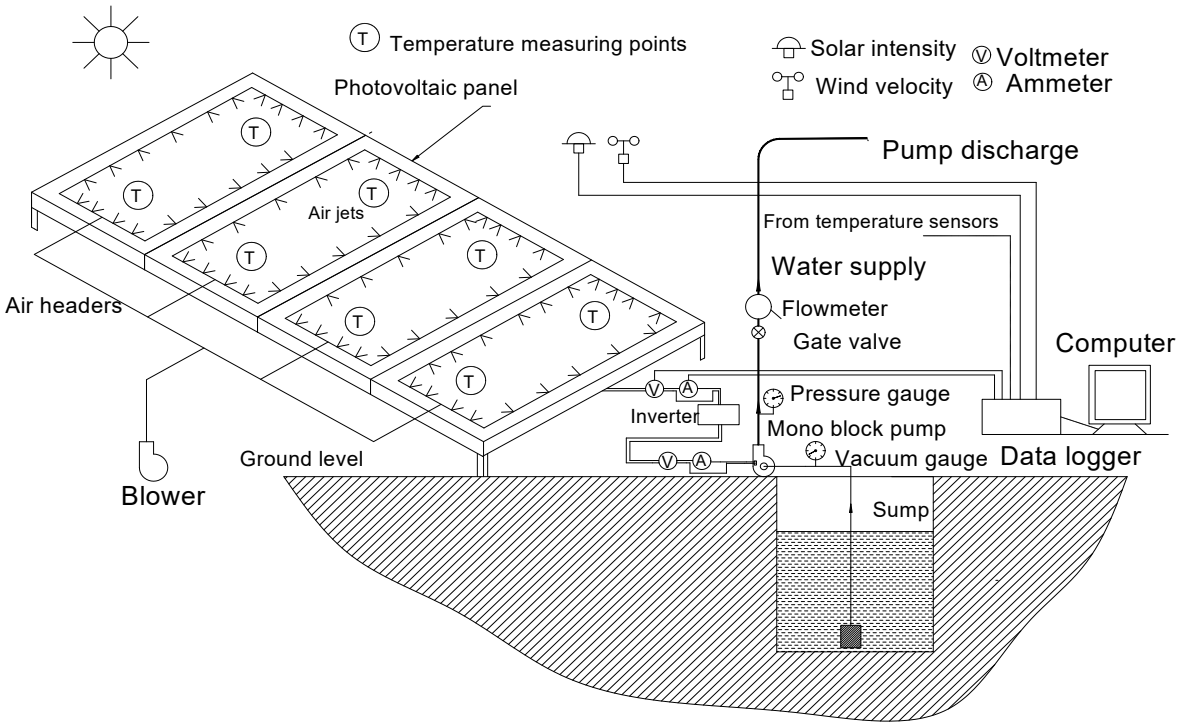


FIG 4.7 Panel air cooling above the panel

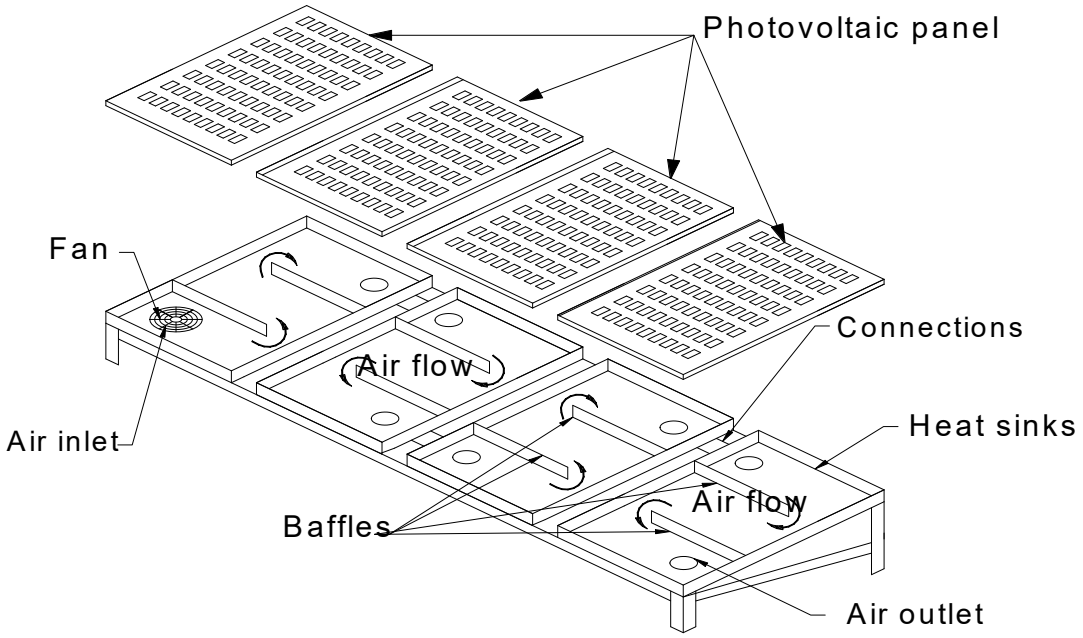


FIG 4.8 Panel air cooling below the panel

CHAPTER 5

ECONOMICAL ASSESSMENT

Pump running time	8 hours per day (during sunshine hours only)
Capital investment	INR. 20,000 (includes equipment and installationcost)
Annual maintenance cost	INR. 1000 (for first five years) and INR1250 (for second five years). (Manufacturer recommendations)
Energy cost per kWh	INR 10.00 (Commercial tariff) from TNEB
Pay back period	6.64 years
Return on Investment	15%
Net present value after 10 yrs.	INR 1,02,986

TABLE 1.4

Specifications and accuracy of measuring instruments

Instrumentation	**Specification (Range)**	**Accuracy**
Rotameter	0-100 LPM	± 1 LPM
Temperature sensor	PT 100 (0-200°C)	± 0.2 ^{o}C
Wattmeter	Digital type	± 2 W
Energy meter	Digital	± 0.5%
Pyranometer	Class-A(0-1200 W/m^2)	± 5 W/m^2
Digital temperature indicator	16 channel indicator (0-200°C)	0.1^{O}C RESOLUTION
Tachometer	0-3000 rpm	± 5 RPM

TABLE 1.5

EXPERIMENTAL PROCEDURE

The experimental observations were made at every ten minutes interval during 8.00 am to 6.00 pm. The experiments have been carried out under the meteorological conditions of Coimbatore city in India on clear sunny days with minimum fluctuations of solar radiation during the summer months of 2015. Before conducting experiments with solar assisted photovoltaic panel, the pump was tested using a variable frequency drive at different heads ranging from 5 m to 25 m with the frequency ranges between 30 and 50 Hz. Based on the experimental observations, polynomial equations were developed for predicting the performance of a centrifugal pump at different photovoltaic power outputs. Initially, the experiments were carried in the solar photovoltaic water pumping system without the influence of panel cooling. The performance parameters such as, current, voltage, instantaneous power outputs of photovoltaic power output, pump discharge were experimentally measured. After the baseline reference test, the performance was carried out using air as cooling medium on the above and the beneath surface of the panel. Further, the performance was evaluated using water cooling arrangement on the above and beneath surface of the panel. The water flow rate was optimized for cooling the photovoltaic panel based on series of trial experiments.

The water flow rate used for cooling of photovoltaic panel was controlled using a control valve. Similarly, all the performance parameters were measured experimentally for the other four modes of operation. Experimental observations have been made for the period of one month under each mode. However, the experimental observations with similar ambient variations observed on 24th day of February 2015 (without the influence of panel cooling), 23rd day of March 2015 (cooling above the panel), 21st day of April 2015 (cooling at beneath surface of the panel), 23rd day of June 2015 (cooling above the panel surface), 23rd day of September 2015 (cooling below the panel surface) were considered for performance comparison. During these days, the similar ambient (such as solar irradiation, ambient temperature and wind velocity) variations were observed.

UNCERTAINITY ANALYSIS

The uncertainties in experiments are calculated using the relation given by Holman (2007). In this research work, the voltage, current, power output, power consumed by the motor, water flow rate, solar irradiation, ambient temperature and air velocity over the photovoltaic panel surface are measured using ammeter, voltmeter, Wattmeter, rotameter, pyranometer, thermocouples and cup type anemometer, respectively. The accuracy of measuring instruments is listed in Table 4.3. The uncertainties in measured and calculated parameters are given by Equation (4.1).

$$w_r = \left[\left(\frac{\partial R}{\partial x_1} w_1 \right)^2 + \left(\frac{\partial R}{\partial x_2} w_2 \right)^2 + \ldots\ldots\ldots + \left(\frac{\partial R}{\partial x_n} w_n \right)^2 \right]^{1/2}$$

Here, R is a given function,

wr is the total uncertainty,

$x_1, x_2, \ldots x_n$ is the independent variables,

w1, w2…wnis the uncertainty in the independent variables.

SUMMARY

The details on specifications of solar photovoltaic water pumping system, accuracy of measuring instruments used in experimentation, detailed experimental procedure and uncertainties in experiments were described in this chapter. The experimental results are shown in next chapter.

CHAPTER 6

ENVIRONMENTAL IMPACT ASSESSMENTS

The environmental impacts of solar photovoltaic water pumping system were assessed for the period of twenty years lift time. The environmental impact assessment was made according to ISO 14040 series of standards. In this research work, the electricity consumption for pumping water 1200 litres of water per day for the head of 15 m was considered for performance comparison. The operation time of the pump is 8 hours per day. Due to lack of manufacturing and disposal data of centrifugal pumps, the environmental impact caused during its manufacturing phase was ignored in this research work. The environmental impacts caused during manufacturing and disposal is also not quantified in the literature. Hence, the environmental impact during disposal stage was also ignored in this work. The electricity consumption of 380 W centrifugal pump for pumping water for 8 h duration is about 3.5 kWh. The photovoltaic panel connected with water pumping system is capable of producing about 4.5 kWh of electricity, which fulfils the requirements for water pumping and also for running the accessories. The yearly consumption of electricity is about 1280 kWh/year. The carbon-dioxide emission for consumption of 1280 kWh of electricity was estimated to be about 1150 kg of CO2. It is observed that, it is possible to reduce 1150 kg of CO2 emission by replacing the existing grid connected water pumping systems with solar photovoltaic water pumping systems.

UNCERTAINTY ANALYSIS

The uncertainties in the experimentally measured parameters such as, photovoltaic power output, photovoltaic efficiency, pump discharge, pump efficiency and overall efficiency are calculated as ± 4 W, ± 0.5%, ± 0.1 m3/s and ± 0.5%, respectively.

SUMMARY

The performance of a solar photovoltaic water pumping system under the influence of panel cooling is described in this chapter. The results presented in this chapter confirms that the solar photovoltaic water pumping system is technically, economically and environment friendly option for domestic, industrial, commercial and agricultural applications. The major outcomes of this research are presented in next chapter.

CHAPTER 7

ADVANTAGES AND DISADVANTAGES OFSOLAR WATER PUMP

ADVANTAGES

- It is clean and non polluting
- It is a renewable energy
- Solar cells do not produce noise.
- Less maintenance
- Everlasting source of energy
- No fuel cost
- Flexibility to be implemented in remote locations.

DISADVANTAGES

- Solar power is not available during night.
- Initial cost is more.
- Dust accumulation will affect the performance of the syste

CHAPTER 8

CONCLUSION

The energy performance of a solar photovoltaic water pumping system was evaluated under five conditions such as:

(a) Photovoltaic panel without cooling,
(b) Photovoltaic panel with air cooling over the panels,
(c) Photovoltaic with air cooling at the beneath surface of the panel,
(d) Photovoltaic panel with water cooling above the panel and
(e) Photovoltaic panel with water cooling beneath the panel.

The experimental observations were made under the meteorological conditions of Coimbatore city in India during the year 2015. Mathematical models have been developed for simulating the performance of solar photovoltaic water pumping systems working under five different climatic conditions. The major conclusions arrived in this research work are consolidated in this chapter.

ANALYTICAL MODELING

A mathematical model was developed for predicting the year round performance with reference to solar intensity, ambient temperature and ambient wind velocity. The results obtained from mathematical models are compared with experimentally observed results for validation. The results predicted using mathematical models are found to be closer to the experimental results with

maximum fraction of variance of 0.9999 and minimum root mean square values and coefficient of variance.

Analysis of variance (ANOVA) was used to identify the significance of ambient parameters and pump operating characteristics influencing the performance of the photovoltaic water pumping system. The results showed that solar intensity and pumping head are the most influencing parameters affecting the performance of photovoltaic water pumping system. The ambient wind velocity and ambient temperature has minor influence on the energy performance of solar photovoltaic water pumping systems. The relative humidity has negligible influence on performance. However, the long term performance of photovoltaic water pumping systems is influenced by relative humidity due to the moisture migration intro cell enclosure. Due to the influence of moisture absorption, the cell performance highly affected, which reduced the photovoltaic power output.

PERFORMANCE COMPARISON

The cooling of photovoltaic panels have significantly reduced the cell temperature, which enhances the photovoltaic power output. The cell temperature was maintained in the range between 25-35 oC and 30-40 oC with the help of water cooling and air cooling, respectively. The results of the research investigation confirmed that, water cooling has significant influence on panel temperature when compared to air cooling due to its higher specific heat when compared to air. The cooling arrangement provided over the beneath surface of the panel of panels (using air or water) has significant reduction in panel cell temperature when compared to the panel cooling above the surface. The cooling of photovoltaic panel has improved

the performance of the photovoltaic efficiency, subsystem efficiency (pump discharge efficiency) and total efficiency by about 3.5%, 2.5% and 2%, respectively using air cooling.

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